cost about a billion U.S. dollars a year (which would represent 7 percent of the present U.S. civilian space budget). Humans would arrive only when the base is fully ready. The use of robot construction crews under radio command from Earth is said to reduce the cost tenfold. The only trouble with the scheme, according to reports, is that other scientists in Japan keep asking, "What's it for?" That's a good question in every nation.

The first human mission to Mars is now probably too expensive for any one nation to pull off by itself. Nor is it fitting that such a historic step be taken by representatives of only a small fraction of the human species. But a cooperative venture among the United States, Russia, Japan, the European Space Agency—and perhaps other nations, such as China—might be feasible in the not too distant future. The international space station will have tested our ability to work together on great engineering projects in space.

The cost of sending a kilogram of something no farther away than low Earth orbit is today about the same as the cost of a kilogram of gold. This is surely a major reason we have yet to stride the ancient shorelines of Mars. Multistage chemical rockets are the means that first took us into space, and that's what we've been using ever since. We've tried to refine them, to make them safer, more reliable, simpler, cheaper. But that hasn't happened, or at least not nearly as quickly as many had hoped.

So maybe there's a better way: maybe single-stage rockets that can launch their payloads directly to orbit; maybe many small payloads shot from guns or rocket-launched from airplanes; maybe supersonic ramjets. Maybe there's something much better that we haven't thought of yet. If we can manufacture propellants for the return trip from the air and soil of our destination world, the difficulty of the voyage would be greatly eased.

Once we're up there in space, venturing to the planets, rocketry is not necessarily the best means to move large payloads around, even with gravity assists. Today, we make a few early rocket burns and later midcourse corrections, and coast the rest of the way. But there are promising ion and nuclear/electric propulsion systems by which a small and steady acceleration is exerted. Or, as the Russian space